Mean and variance of a discrete distribution

Aim

To find mean and variance of arrival of objects from the feeder using probability distribution

Software required

Python and Visual components tool

Theory

The expectation or the mean of a discrete random variable is a weighted average of all possible values of the random variable. The weights are the probabilities associated with the corresponding values. It is calculated as,

$$\mathbf{E(X) = \mu = \Sigma_i\, x_i\, p_i \qquad i = 1, 2, \ldots, n}$$

$$E(X) = x_1p_1 + x_2p_2 + \ldots + x_np_n.$$

The variance of a random variable shows the variability or the scatterings of the random variables. It shows the distance of a random variable from its mean. It is calcualted as

$$\sigma_x^2 = \mathrm{Var}\,(X) = \sum_i (x_i - \mu)^2\, p(x_i) = E(X - \mu)^2 \text{ or, } \mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

$$E(X^2) = \sum_i x_i^2\, p(x_i), \text{ and } [E(X)]^2 = [\sum_i x_i\, p(x_i)]^2 = \mu^2.$$

Procedure

1. Construct frequency distribution for the data
2. Find the probability distribution from frequency distribution.
3. Calculate mean using

$$\mathbf{E(X) = \mu = \Sigma_i\, x_i\, p_i \qquad i = 1, 2, \ldots, n}$$

4. Find

$$E(X^2) = \sum_i x_i^2\, p(x_i).$$

5. Calculate variance using

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

Experiment

Program

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Declaring the value of n

```
import numpy as np
n = int(input("Enter the value of n : "))
print("Value of n =", n)
```

Getting the inputs

```
InputVal = {}
for i in range(1, n+1):
    val = int(input(f"Enter the value no {i} : "))
    try:
```

```
        InputVal[val] += 1
    except:
        InputVal[val] = 1
print(f"{i} Values Collected Successfully")
```

Finding Mean

```
mean = 0
for key, val in InputVal.items():
    mean += key*(val/n)
print(f"Mean = {mean:.3f}")
```

Finding Variance

```
ex2 = 0
for key, val in InputVal.items():
    ex2 += ((key**2) * val/n)
var = ex2 - mean**2
print(f"Variance : {var:.3f}")
```

Finding Standard Deviation

```
from math import sqrt
sdtDeviation = sqrt(var)
print(f"Standard Deviation = {sdtDeviation:.3f}")
```

Output

Refer to the following images to view the output of the program

Declaring the value of n

```
import numpy as np
n = int(input("Enter the value of n : "))
print("Value of n =", n)
```

```
Value of n = 10
```

Getting the inputs

```
InputVal = {}
for i in range(1, n+1):
    val = int(input(f"Enter the value no {i} : "))
    try:
        InputVal[val] += 1
    except:
        InputVal[val] = 1
print(f"{i} Values Collected Successfully")
```

```
10 Values Collected Successfully
```

Finding Mean

```
mean = 0
for key, val in InputVal.items():
    mean += key*(val/n)
print(f"Mean = {mean:.3f}")
```

```
Mean = 2.800
```

Finding Variance

```
ex2 = 0
for key, val in InputVal.items():
    ex2 += ((key**2) * val/n)
var = ex2 - mean**2
print(f"Variance : {var:.3f}")
```

```
Variance : 4.960
```

Finding Standard Deviation

```
from math import sqrt
sdtDeviation = sqrt(var)
print(f"Standard Deviation = {sdtDeviation:.3f}")
```

```
Standard Deviation = 2.227
```

Result

The mean and variance of arrivals of objects from feeder using probability distribution are calculated.